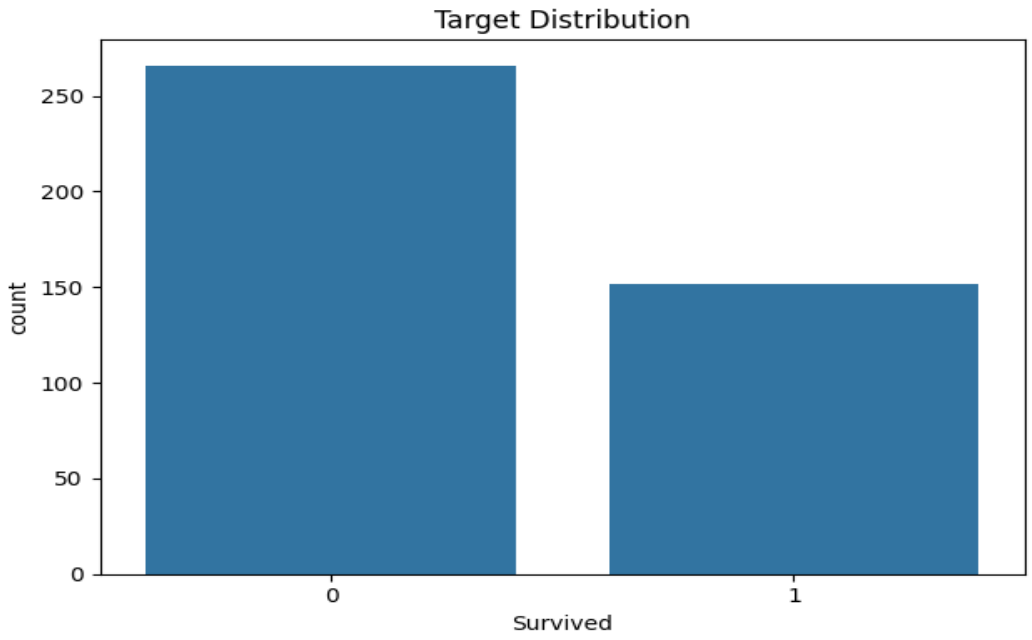
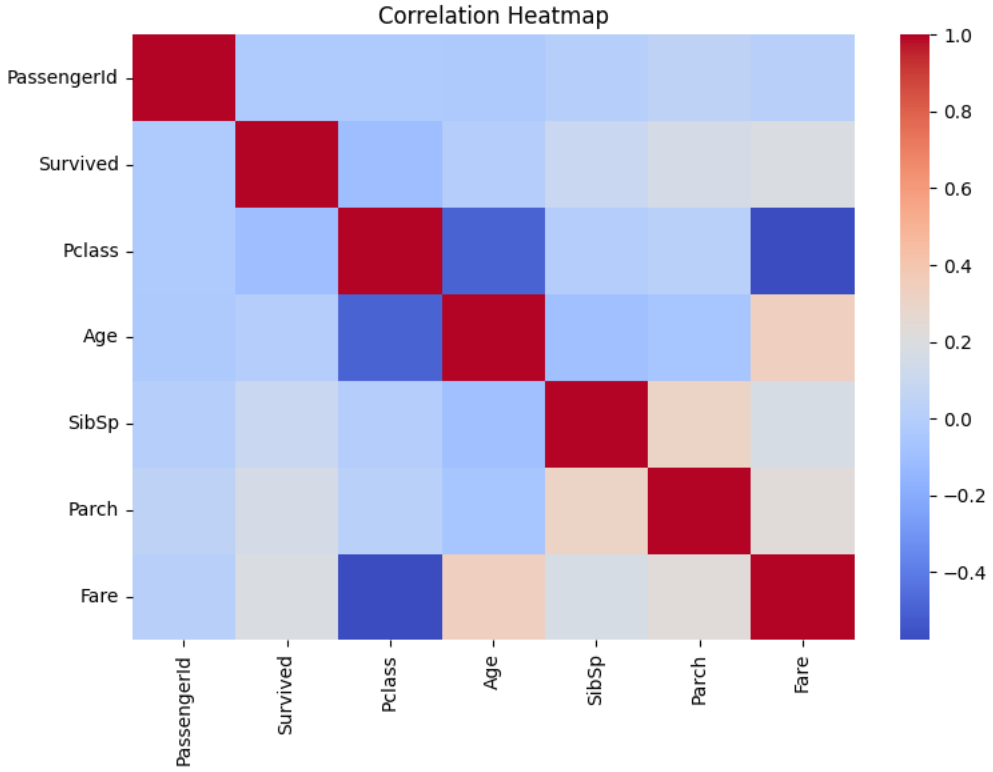
AutoML Model Report

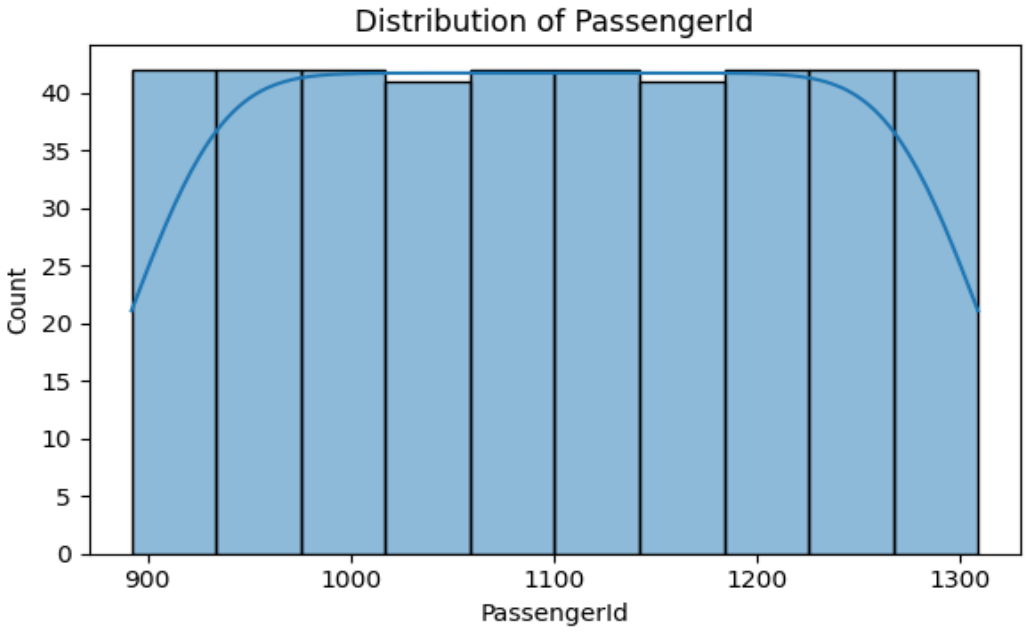
Dataset Overview

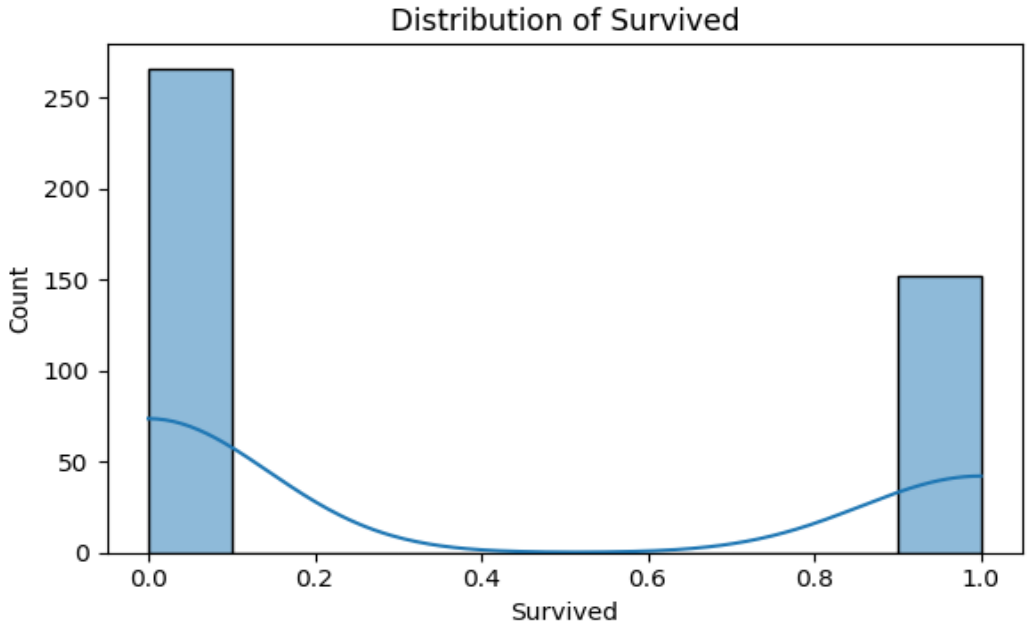
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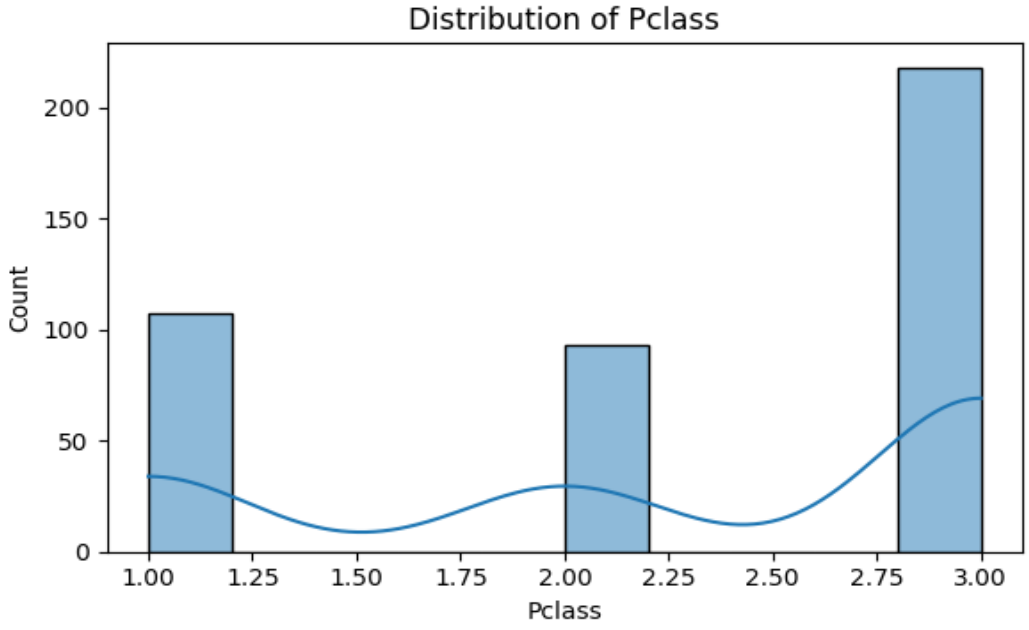
Exploratory Data Analysis

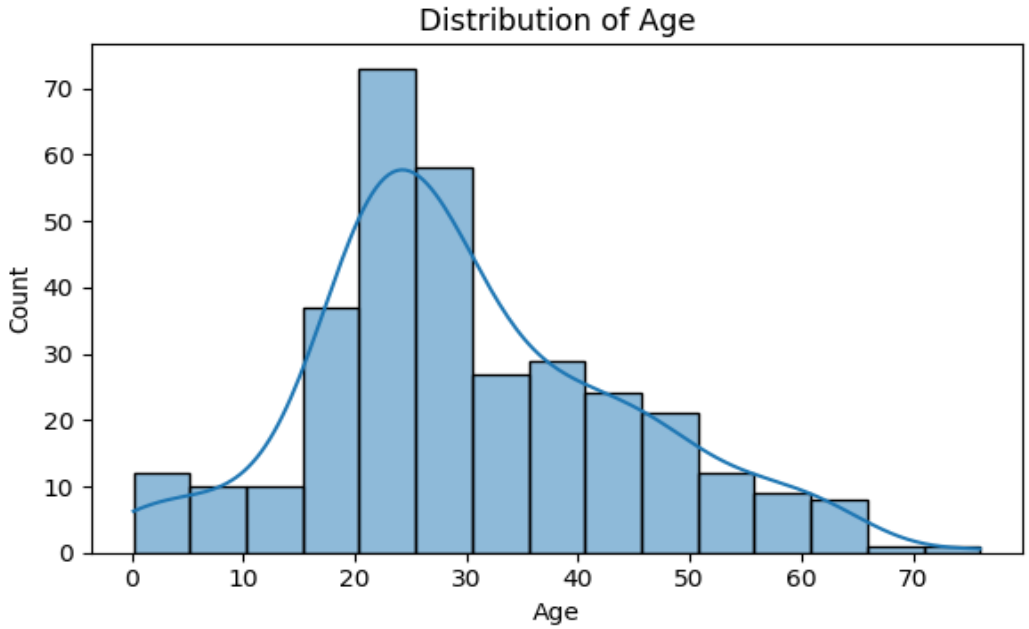


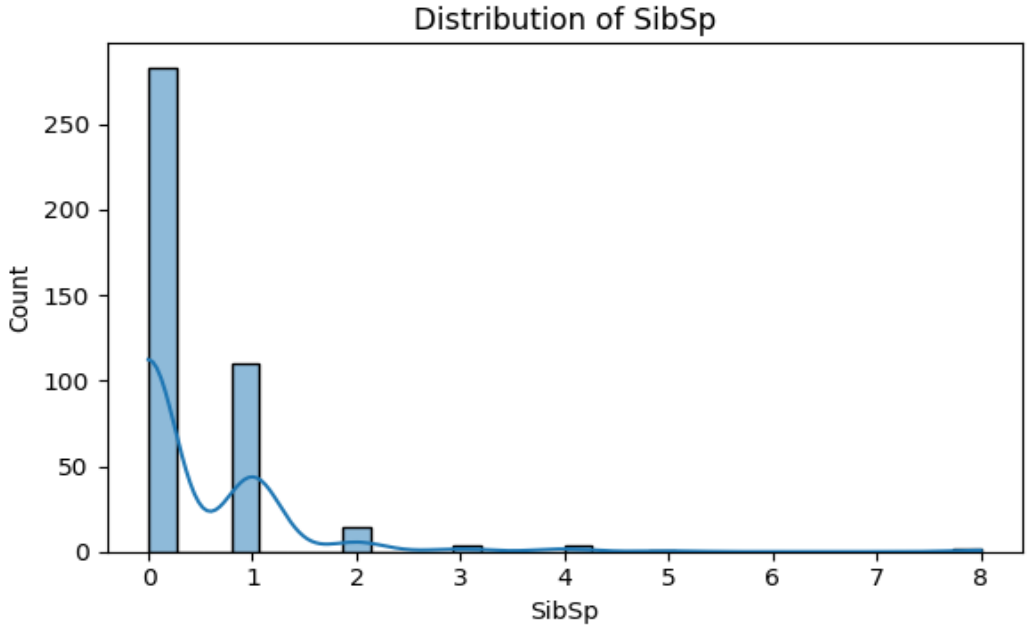

Correlation Heatmap
PassengerId
Survived
Pclass
Age
SibSp
Parch
Fare
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4

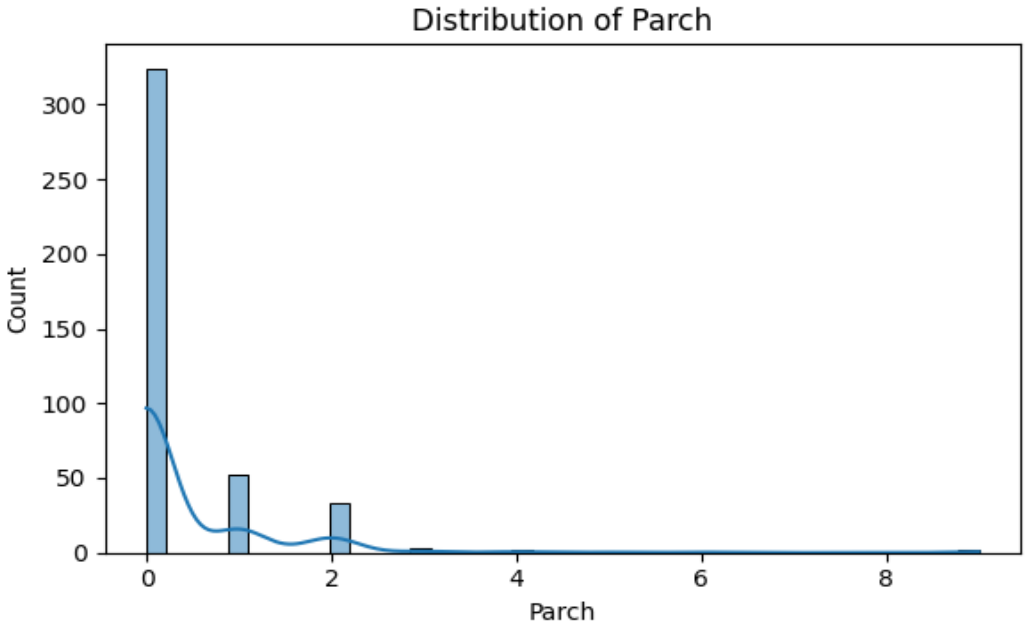

Distribution of PassengerId
Count
0
5
10
15
20
25
30
35
40
900
1000
1100
1200
1300
PassengerId


Distribution of Survived
Count
250
200
150
100
50
0
0.0
0.2
0.4
0.6
0.8
1.0
Survived


Distribution of Pclass
Count
200
150
100
50
0
1.00
1.25
1.50
1.75
2.00
2.25
2.50
2.75
3.00
Pclass


Distribution of Age
Count
Age
0
10
20
30
40
50
60
70


Distribution of SibSp
Count
SibSp
0
50
100
150
200
250
0
1
2
3
4
5
6
7
8

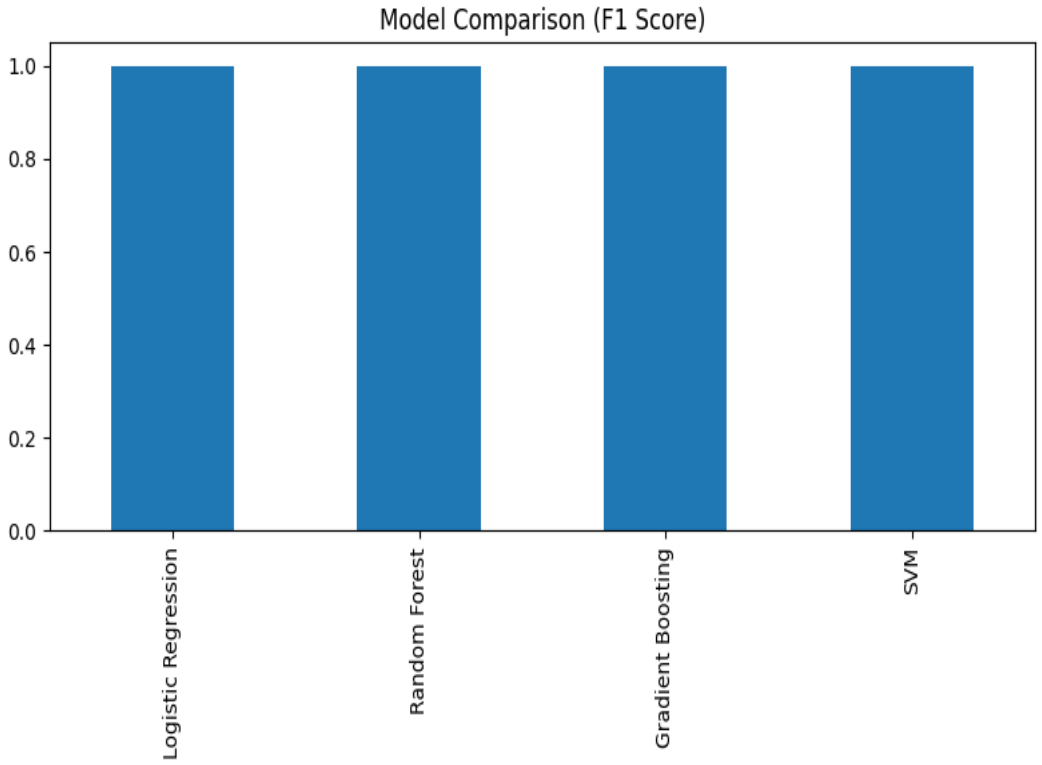

Distribution of Parch
Count
300
250
200
150
100
50
0
0
2
4
6
8
Parch

Data Preprocessing

Age: filled_mean
Fare: filled_mean
Cabin: dropped_high_missing

Feature Selection

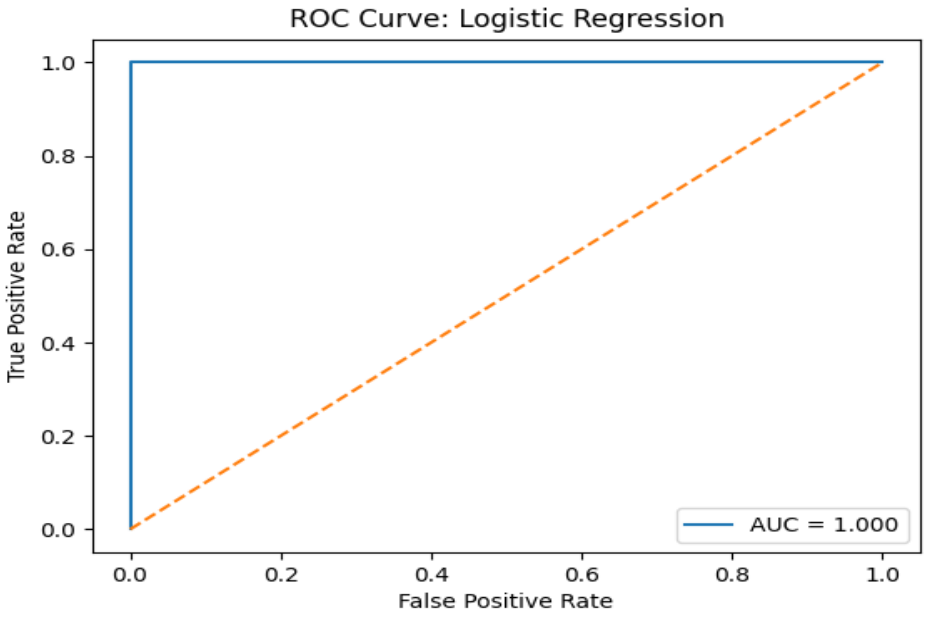
Model Comparison

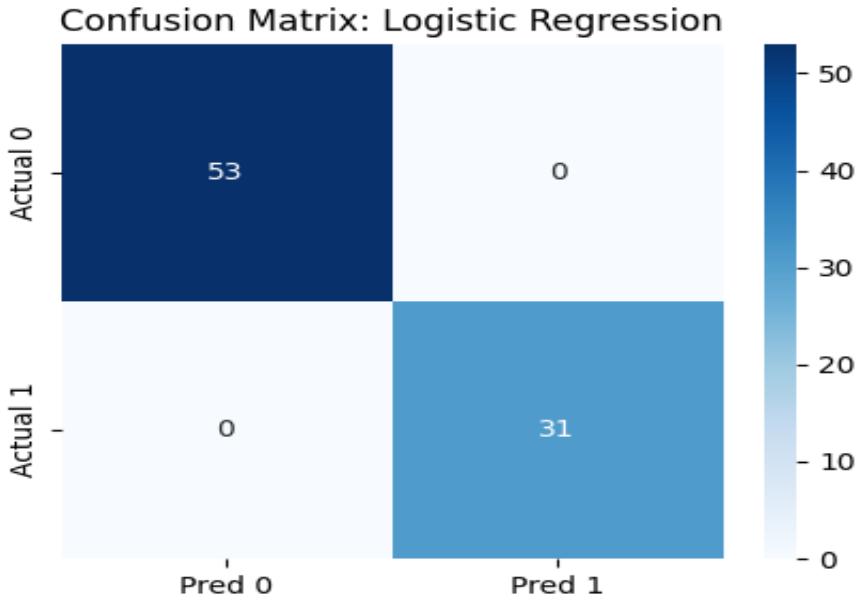


Model	Accuracy	Precision	Recall	F1
Logistic Regression	1.000	1.000	1.000	1.000
Random Forest	1.000	1.000	1.000	1.000
Gradient Boosting	1.000	1.000	1.000	1.000
SVM	1.000	1.000	1.000	1.000

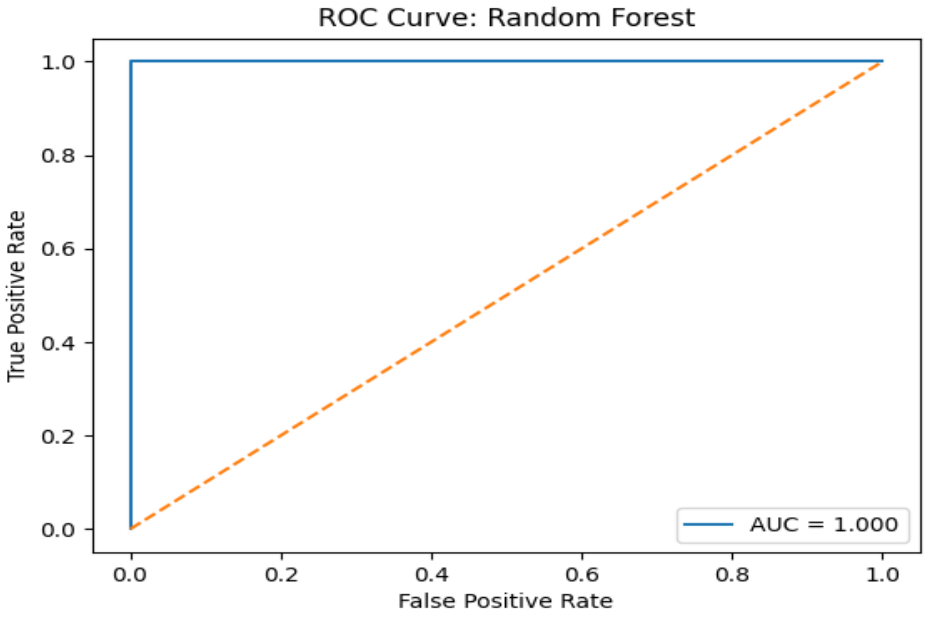
Individual Model Performance

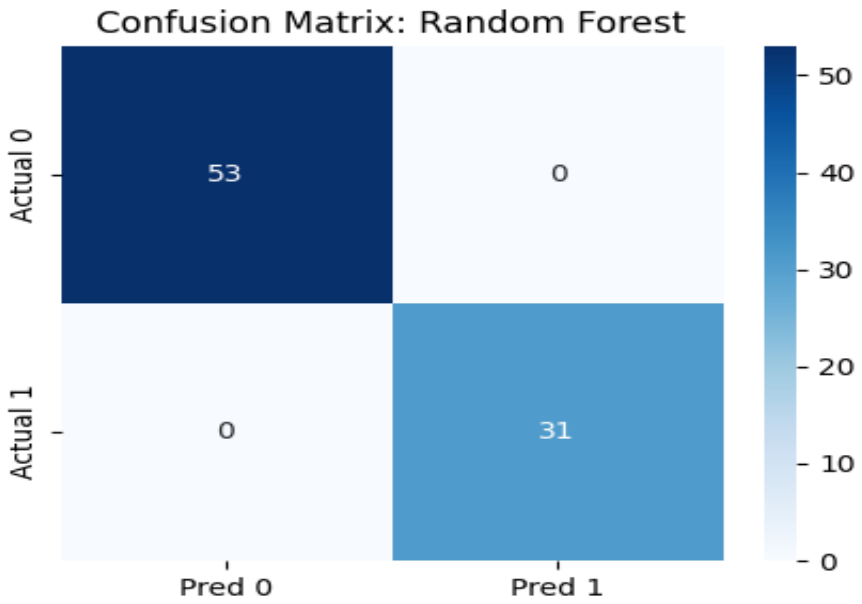
Logistic Regression



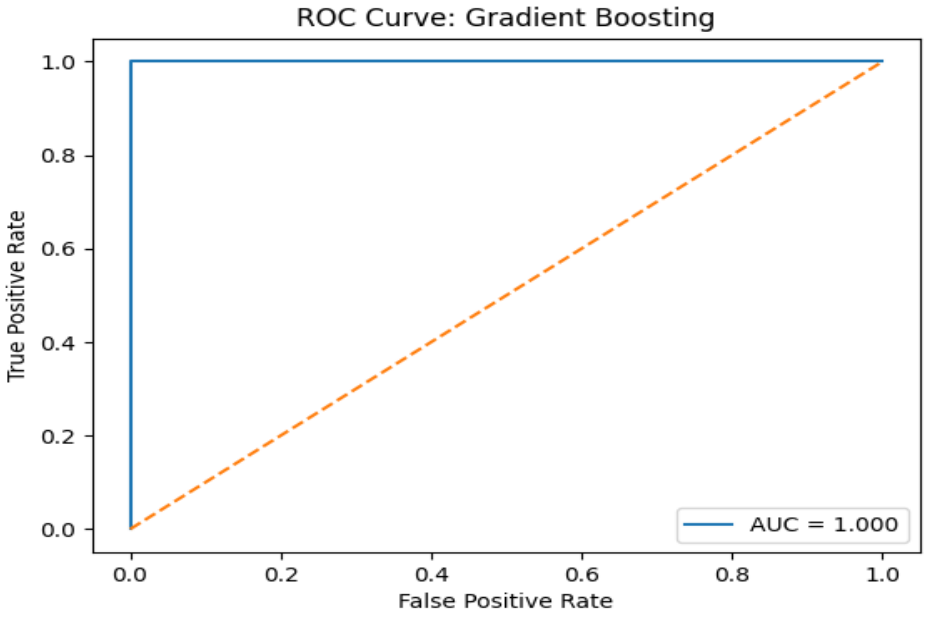


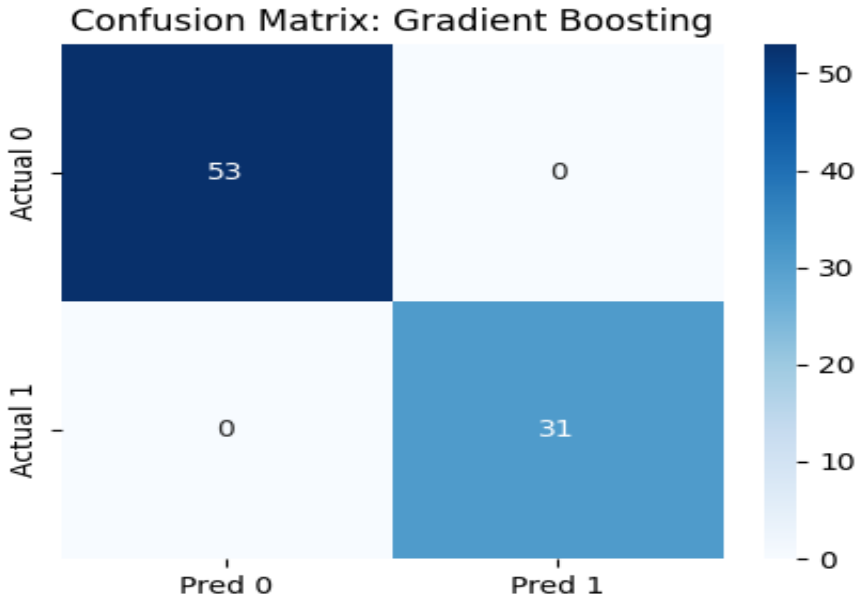
Random Forest



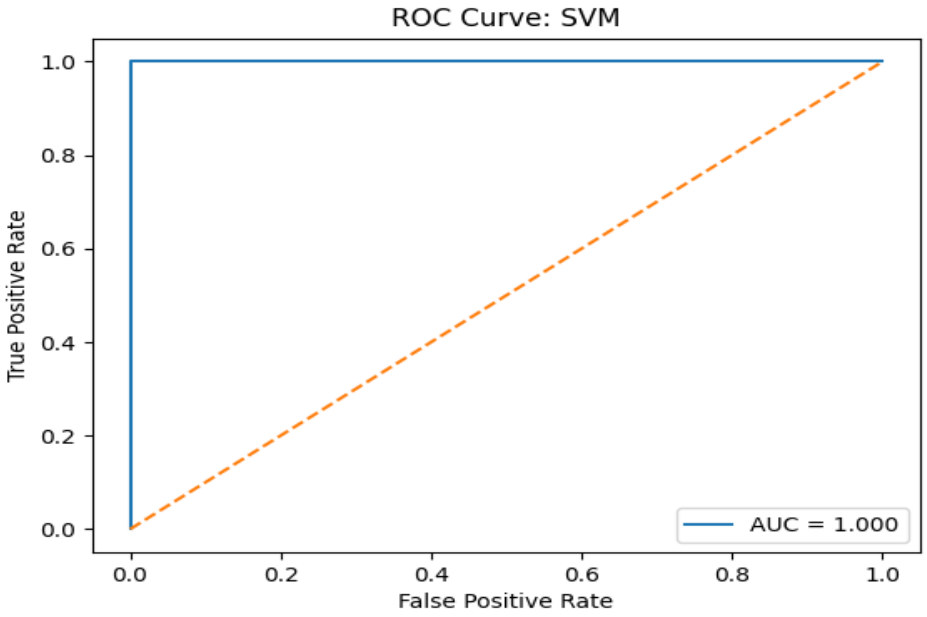


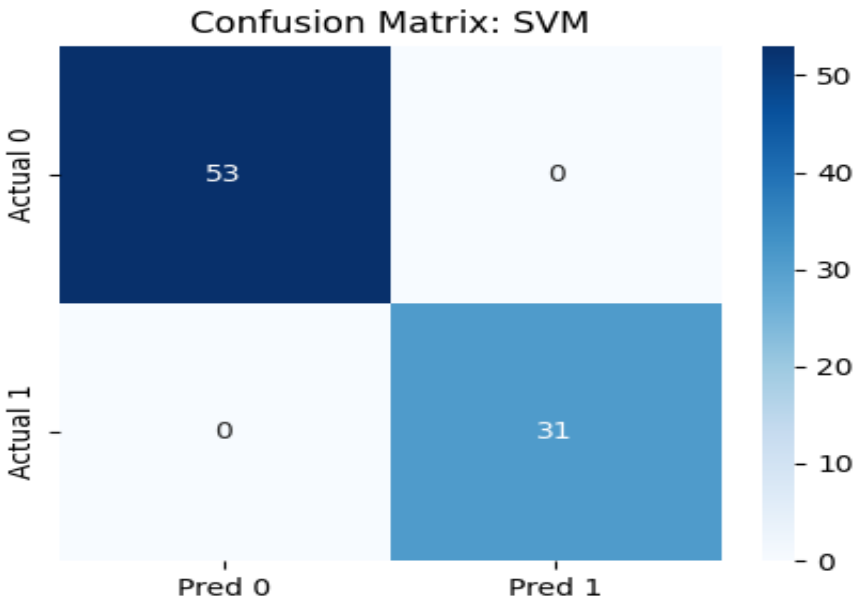
Gradient Boosting


ROC Curve: Gradient Boosting
True Positive Rate
False Positive Rate
AUC = 1.000
0.0
0.2
0.4
0.6
0.8
1.0


Confusion Matrix: Gradient Boosting
Actual 0
Actual 1
Pred 0
Pred 1
53
0
0
31
50
40
30
20
10
0

SVM


ROC Curve: SVM
True Positive Rate
False Positive Rate
AUC = 1.000
0.0
0.2
0.4
0.6
0.8
1.0


Confusion Matrix: SVM
Actual 0
Actual 1
Pred 0
Pred 1
53
0
0
31
50
40
30
20
10
0

Best Model Analysis

The best model selected by the AutoML pipeline is: **Logistic Regression**